

Real World Case Study Paper

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Introduction

The corporation that will be the subject of this case study for effective cybersecurity practices will be Sony Interactive Entertainment (SIE). SIE is a large multinational company with geographically distributed infrastructure; however, they are not independently operated and in fact exist under the umbrella of its parent company, the Sony Group Corporation, which has several different international divisions. This is important to note because the cybersecurity practices and implementation methods utilized by SIE are most certainly subject to influence from the policies, priorities, and resource constraints of its parent corporation and the needs of the other branch companies. However, for the sake of scope for this case study, their best practices, cybersecurity issues, and possible methodologies to implement solutions will be examined independently. SIE is responsible for the management and production of Sony's digital entertainment products and hardware. This includes the digital entertainment media from PlayStation Studios (PSS), their hardware division for the PlayStation console, virtual reality, and handheld lineup, and their online streaming services that are supported by the networking infrastructure that serves their userbase. The purpose of this case study is to examine the security practices and deficiency implications of those online services and determine possible solutions and their impact. Over its years of operations, SIE's online network has been infamous for massive cybersecurity related flaws in the industry that have led to historic data breaches coupled with disastrous mismanagement in the media.

Best Practices

To begin, now that the scope of this case study is clearly defined, we can determine the best practices that SIE is currently implementing for their security standards and later highlight

examples of issues that have occurred. Their online services operate from and utilize the IT infrastructure of PlayStation Network (PSN). PSN supports a complex array of features, from being the backbone for the online multiplayer implemented by the various developers on the platform to SIE's online marketplace that handles all financial transactions and supports their live streaming services. These features are used and relied on by over 105 million active users worldwide. (SGP, 2023). From a business continuity perspective, one of the areas in which Sony is implementing best practices is the resiliency of PSN infrastructures. There are 11 primary server clusters that centralize and manage traffic among the different regions of PSN's geographically distributed userbase. (Kovacs, 2023). This benefits the network in several ways, as the overall distributed nature of the IT infrastructure makes it better at handling demand spikes and is more resistant to natural disasters and better at disaster recovery. A relatively recent innovation towards maintaining high availability on their services is the integration of AWS cloud infrastructure to increase scalability and flexibility in the network for changing demands (SIE Case Study, 2022). These advancements help to ensure principles like the recovery time objective (RTO), which is the amount of time an organization can tolerate for recovering to normal operations, are maintained as the hardware infrastructure is resistant to incidents.

PSN handles a variety of financial transactions initiated based on the features and services that SIE offers, highlighted earlier. On average the network handles 50 million software sales in a single business quarter. (SGP, 2023) This doesn't even include the variety of annual and monthly service membership programs that utilize and store personally identifiable information (pii) such as names, addresses, and phone numbers, as well as financial information like credit card numbers. As a result, securing this data is paramount to maintaining confidentiality on the network. This is accomplished through two main focuses, data security and

user domain best practices. Sony both encrypts stored data on PSN, while offering end-to-end encryption methods for data in transit from communication between their client hardware and the network. (Kovacs, 2023). Account creation policies also follow best practices in the user domain from implementing additional access control features like two-factor authentication. Also, geolocation is inherently baked into account management making these details unexploitable by external threat agents. (PlayStation, 2023). AI based software also automatically flags accounts with suspicious activity in real-time is an implementation of automation and monitoring that helps contain compromised accounts and reduce overall risk.

Problems and Solutions

A multinational, large scale corporation has a significant budget for security practices and implementation to protect and harden their infrastructure and assets. However, that does not make them immune to cybersecurity issues and faults. The three main ethical, legal, and cybersecurity related problems of this case study for SIE will revolve around: SIE'S data collection policies with respect to their data breaches and confidentiality concerns, service outages and network availability issues from cyberattacks, and disaster recovery management from their information PR policies towards the public regarding these issues. Methodologies that could implement possible mitigation solutions will be proposed for each of these problems.

The first major issue to address is availability concerns from cyberattacks and service interruptions. As noted earlier, PSN supports a wide variety of business-critical services. Therefore, outages on this infrastructure can be very detrimental. Although sufficiently hardened against natural disasters, one of the largest interruptions in the network's history was a distributed denial of service (DDoS) attack perpetrated in 2014, affecting nearly 160 million

users. (Reed, 2023). A possible solution to this that SIE has even invested in, in recent years, is the updating of legacy IT infrastructure with more flexible cloud solutions. This allows for better system support and load balancing for resilience against outages. Additionally, automated monitoring services for traffic management and load spikes are an additional benefit. (SIE Case Study, 2022).

For more than a decade SIE has been the target of several severe data breaches in the gaming industry. The most egregious of which was a series of breaches in 2011 that saw pii and financial information exposed for nearly 77 million PSN users. This is sensitive information was stored without encryption measures and entangled in directories with other identifying information and financial data. (The Guardian, 2011) Additionally, this attack went undetected for a several days before Sony was able to identify that it had been breached. This track record is not ideal when PSN has been documented logging substantial amounts of sensitive personal information for each of its users and even caching that data for significant periods of time. (The Guardian, 2022) Data breaches and mismanagement of data are serious security risks that can be mitigated with adequate management practices and proper data classification. Being able to determine what data is deemed sensitive and what data on the network could be at risk for exposure. Extensive classification would help maintain standards of data compliance as well. For example, Payment Card Industry Data Security Standards (PCI-DSS) require certain levels of data security when handling payment information and ensuring that these standards are being followed can minimize the exposure of sensitive data in cyberattacks. (Spirion,2023).

The third issue that we will propose a solution for is SIE incident management handling and PR actions. This is primarily from an ethical perspective, as in the data breaches and security incidents mentioned in the other issues, SIE PR often delayed communication and hid necessary

details to users about these incidents (The Guardian, 2011). This can cause significant issues as users were not aware that their data had been exposed and the proper steps hadn't been taken to protect their assets until several days after the initial incident. Poor PR management exacerbates security concerns with both users and stakeholders. Strategies to correct this issue involve implementing and revamping communication frameworks for incident response measures. A chief information officer (CIO) would set the drive for response measures and be able to push a unified direction and message both internally and externally. Ultimately, the primary cause for their PR management would be difficult to find, but it can generally track back to a lack of communication between the technical teams handling Sony's security incidents and their PR departments.

Conclusion

Overall, like any large corporation in the modern world, SIE suffers from security issues that have only become more prevalent to solve in the digital age. Properly securing users personal and financial information against cyberthreats and preventing data leaks. Maintaining a resilient IT infrastructure to keep their business-critical services and communications up for millions of users concurrently. As well as clear and concise public communication to help foster trust in the company's ability to protect and secure users' information. The solutions suggested in would go to mitigate these issues and help maintain security best practices. Some examples of SIE already doing this include the as mentioned AWS cloud integration on their infrastructure, the lack of any major service outages in recent years of operation, and more concise PR communication standards for recent incidents. (Infotech, 2024)

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